

Algebra & Equations

ANSWER KEY



Solving linear equations

- 1 Solve $3x - 7 = 11$.
 $3x = 18$, so $x = 6$.
 - 2 Solve $5(x - 2) = 3x + 4$.
 $5x - 10 = 3x + 4 \Rightarrow 2x = 14 \Rightarrow x = 7$.
 - 3 Solve $\frac{3x - 2}{4} = x - 1$.
 $3x - 2 = 4(x - 1) = 4x - 4 \Rightarrow -x = -2 \Rightarrow x = 2$.
 - 4 Solve $2(x + 3) - 5 = 3(x - 1)$.
 $2x + 1 = 3x - 3 \Rightarrow -x = -4 \Rightarrow x = 4$.
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Expanding and simplifying expressions

- 5 Expand and simplify $3(2x - 1) - 2(x - 4)$.
 $6x - 3 - 2x + 8 = 4x + 5$.
 - 6 Expand $(x + 5)(x - 2)$.
 $x^2 - 2x + 5x - 10 = x^2 + 3x - 10$.
 - 7 Expand $(2x - 3)^2$.
 $(2x - 3)^2 = 4x^2 - 12x + 9$.
 - 8 Simplify $\frac{6x^2 - 9x}{3x(2x - 3)}$ (for $x \neq 0$).
 $\frac{3x(2x - 3)}{3x} = 2x - 3$.
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Factoring quadratic expressions

- 9 Factor $x^2 + 7x + 12$.
 $(x + 3)(x + 4)$.
- 10 Factor $x^2 - 2x - 15$.
 $(x - 5)(x + 3)$.
- 11 Factor $2x^2 + 5x - 3$.
 $(2x - 1)(x + 3)$.

- 12 Factor $x^2 - 49$ using the difference of squares.
 $(x - 7)(x + 7)$.
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Solving quadratic equations by factoring

- 13 Solve $x^2 - 5x + 6 = 0$.
 $(x - 2)(x - 3) = 0$, so $x = 2$ or $x = 3$.
- 14 Solve $x^2 + 2x - 8 = 0$.
 $(x + 4)(x - 2) = 0$, so $x = -4$ or $x = 2$.
- 15 Solve $2x^2 - 5x - 3 = 0$.
 $(2x + 1)(x - 3) = 0$, so $x = -\frac{1}{2}$ or $x = 3$.
- 16 Solve $x^2 - 9x + 20 = 0$.
 $(x - 4)(x - 5) = 0$, so $x = 4$ or $x = 5$.
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The quadratic formula and the discriminant

- 17 Use the quadratic formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ to solve $x^2 + 3x + 1 = 0$, leaving the answer in exact form.
 $a = 1, b = 3, c = 1$. $x = \frac{-3 \pm \sqrt{9 - 4}}{2} = \frac{-3 \pm \sqrt{5}}{2}$.
- 18 Find the discriminant of $2x^2 - 4x + 5 = 0$ and state how many real roots it has.
 $\Delta = (-4)^2 - 4(2)(5) = 16 - 40 = -24$. Since $\Delta < 0$, there are no real roots.
- 19 Find the discriminant of $x^2 - 6x + 9 = 0$ and state how many real roots it has.
 $\Delta = (-6)^2 - 4(1)(9) = 36 - 36 = 0$. Since $\Delta = 0$, there is one repeated real root ($x = 3$).
- 20 Use the quadratic formula to solve $3x^2 - 2x - 1 = 0$.
 $\Delta = (-2)^2 - 4(3)(-1) = 4 + 12 = 16$. $x = \frac{2 \pm 4}{6}$, so $x = 1$ or $x = -\frac{1}{3}$.
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Absolute value equations

- 21 Solve $|x - 4| = 9$.
 $x - 4 = 9 \Rightarrow x = 13$, or $x - 4 = -9 \Rightarrow x = -5$.
- 22 Solve $|2x + 3| = 7$.
 $2x + 3 = 7 \Rightarrow x = 2$, or $2x + 3 = -7 \Rightarrow x = -5$.

- 23 Solve $3|x - 1| = 12$.
 $|x - 1| = 4$, so $x - 1 = 4 \Rightarrow x = 5$, or $x - 1 = -4 \Rightarrow x = -3$.
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Linear inequalities

- 24 Solve $2x - 5 < 9$ and graph the solution set on a number line.
 $2x < 14 \Rightarrow x < 7$.
- 25 Solve $-3x + 4 \geq -8$.
 $-3x \geq -12$. Dividing by -3 reverses the inequality: $x \leq 4$.
- 26 Solve the compound inequality $5 < 2x + 1 \leq 11$.
Subtract 1: $4 < 2x \leq 10$. Divide by 2: $2 < x \leq 5$.
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Systems of linear equations

- 27 Solve the system $x + y = 10$, $x - y = 2$.
Adding the equations: $2x = 12 \Rightarrow x = 6$. Then $y = 10 - 6 = 4$.
- 28 Solve the system $3x + 2y = 16$, $x - y = 2$.
From the second equation, $x = 2 + y$. Substituting: $3(2 + y) + 2y = 16 \Rightarrow 6 + 5y = 16 \Rightarrow y = 2$. Then $x = 4$.
- 29 This question has two parts. The sum of two numbers is 15 and their difference is 3.
- a Write a system of two equations for the numbers x and y . $x + y = 15$ and $x - y = 3$.
- b Solve the system to find both numbers. Adding the equations: $2x = 18 \Rightarrow x = 9$. Then $y = 15 - 9 = 6$.
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Rational equations

- 30 Solve $\frac{x}{x-2} = 3$.
 $x = 3(x-2) = 3x - 6 \Rightarrow -2x = -6 \Rightarrow x = 3$.
- 31 Solve $\frac{2}{x+1} = \frac{1}{3}$.
Cross-multiplying: $6 = x + 1 \Rightarrow x = 5$.
- 32 Solve $\frac{x+1}{x-1} = 2$.
 $x + 1 = 2(x - 1) = 2x - 2 \Rightarrow -x = -3 \Rightarrow x = 3$.
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Solving for a variable in a formula

33 The area of a triangle is $A = \frac{1}{2}bh$. Solve for h .
$$h = \frac{2A}{b}.$$

34 The circumference of a circle is $C = 2\pi r$. Solve for r .
$$r = \frac{C}{2\pi}.$$

35 The formula $v = v_0 + at$ gives velocity under constant acceleration. Solve for t .
$$t = \frac{v - v_0}{a}.$$

Linear-quadratic systems

36 Solve the system $y = x^2 - 1$, $y = 2x + 2$.
Setting equal: $x^2 - 1 = 2x + 2 \Rightarrow x^2 - 2x - 3 = 0 \Rightarrow (x - 3)(x + 1) = 0$. So $x = 3$, $y = 8$ or $x = -1$, $y = 0$.

37 Solve the system $y = x^2$, $y = 3x - 2$.
Setting equal: $x^2 = 3x - 2 \Rightarrow x^2 - 3x + 2 = 0 \Rightarrow (x - 1)(x - 2) = 0$. So $x = 1$, $y = 1$ or $x = 2$, $y = 4$.

Word problems: setting up and solving equations

38 This question has two parts. A rectangular garden has a length that is 4 meters more than its width w . Its area is 96 square meters.

a Write an equation relating w to the area. $w(w + 4) = 96$, i.e. $w^2 + 4w - 96 = 0$.

b Solve for w , and state the garden's length. Factoring: $(w - 8)(w + 12) = 0$. Since width is positive, $w = 8$ meters, so the length is $w + 4 = 12$ meters.

39 This question has two parts. A mobile-data plan charges a fixed monthly fee f plus a rate r per gigabyte used. One month, 5 GB of data cost a total of 65 dollars; another month, 9 GB cost a total of 93 dollars.

a Write a system of two equations for f and r . $f + 5r = 65$ and $f + 9r = 93$.

b Solve the system to find the fixed fee and the rate per gigabyte. Subtracting the first equation from the second: $4r = 28 \Rightarrow r = 7$. Then $f = 65 - 5(7) = 30$. Fixed fee: 30 dollars; rate: 7 dollars per GB.